

Definitions: 9/7

$$A_t(n) = |\{\text{partial sums of length } t\}|$$

$$A_n(n) = A(n)$$

$$A_t(t+1) = A_{t+1}(t+1) \text{ AND } A_{t-1}(t) = A_t(t)$$

$$C_t(n) = A_t(n) - (2n-1)^n$$

CONJECTURES

$$A_t(n) \leq 2^t \cdot n^t \leq (n!)^2 \quad A_0(n) = 1$$

$$\deg(C_t(n)) = n-2 \geq 0 \quad A_1(n) = 2n-1$$

$$C_2(n) = 6$$

$$A_2(n) = 4n^2 - 4n - 5$$

$$C_3(n) = -(36n - 2)$$

$$A_3(n) = 8n^3 - 12n^2 - 36n + 1$$

$$C_4(n) = -(144n^2 + 16n + 410)$$

Pattern in 1st Coefficient?

$$\text{Approximate } A_n(n) = \sum_{i=1}^n (2i-1)^2 - C_n(n) + C_n^{(2)}(n) \dots$$